UNIT 4 CATALYSIS, LP 3 4.5 TYPES OF CATALYSIS

Types of Catalysis

In addition to homogenous and heterogeneous catalysis, the following are the different types

Positive Catalysis

When the rate of the reaction is accelerated by the foreign substance, it is said to be a **positive catalyst** and phenomenon as **positive catalysis.** Examples of positive catalysis are given below.

(i) Catalyst helps in altering the velocity of the reaction and the catalyst is called as positive catalyst. eg. MnO_2 acts as catalyst in decomposition of $KClO_3$ into KCl and O_2. Decomposition of $KClO_3$

$$2\,KClO_3 \xrightarrow[370^\circ C]{MnO_2\,(S)} 2\,KCl + 3O_2$$

Negative Catalysis

There are certain, substance which, when added to the reaction mixture, retard the reaction rate instead of increasing it. These are called negative catalyst or inhibitors and the phenomenon is known as negative catalysis.

Some examples are as follows.

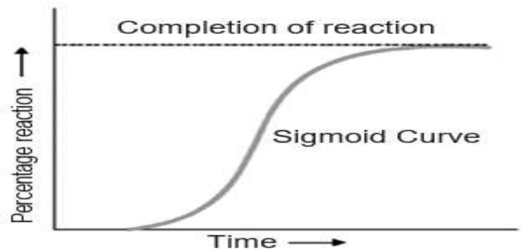
- Oxidation of sodium sulphite
- $$2\,Na_2SO_3 + O_2 \xrightarrow{Alcohol} 2\,Na_2SO_4$$
- Tetra Ethyl Lead (TEL) is added to petrol to retard the ignition of petrol vapours on compression in an internal combustion engine and thus minimize the **knocking effect.**
- Alcohol retards the conversion of chloroform to phosgene

Autocatalysis

When one of the products of reaction itself acts as a catalyst for that reaction the phenomenon is called Autocatalysis.

In autocatalysis the initial rate of the reaction rises as the catalytic product is formed, instead of decreasing steadily (Figure). The curve plotted between reaction rate and time shows a maximum when the reaction is complete

A chemical reaction is said to have undergone autocatalysis, or be autocatalytic, if the reaction product is itself the catalyst for that reaction.



Rate of autocatalytic reaction

Examples of Autocatalysis

- **Hydrolysis of an Ester.** The hydrolysis of ethyl acetate forms acetic acid (CH_3COOH) and ethanol. Of these products, acetic acid acts as a catalyst for the reaction.

$$CH_3COOC_2H_5 + H_2O \longrightarrow \underset{\text{catalyst}}{CH_3COOH} + C_2H_5OH$$

- **Oxidation of Oxalic acid**. When oxalic acid is oxidised by acidified potassium permanganate, manganous sulphate produced during the reaction acts as a catalyst for the reaction.

$$2\,KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow \underset{\text{catalyst}}{2\,MnSO_4} + K_2SO_4 + 8\,H_2O + 10\,CO_2$$

- **Decomposition of Arsine.** The free arsenic produced by the decomposition of arsine (AsH_3) Auto-catalyses the reaction.

$$2\,AsH_3 \longrightarrow \underset{\text{catalyst}}{2\,As} + 3H_2$$

Promoters

The activity of a catalyst can often be increased by addition of a small quantity of a second material. This second substance is either not a catalyst itself for the reaction or it may be a feeble catalyst.

A substance which, though itself not a catalyst, promotes the activity of a catalyst is called a promoter. They are substances when added in small concentration can increase the activity of a catalyst.

Example of Promoters

Molybdenum (Mo) or aluminium oxide (Al_2O_3) promotes the activity of iron catalyst in the Haber synthesis for the manufacture of ammonia.

$$N_2 + 3H_2 \xrightarrow[Mo]{Fe} 2NH_3$$

In some reactions, mixtures of catalysts are used to obtain the maximum catalytic efficiency. For example, in the synthesis of methanol (CH_3OH) from carbon monoxide and hydrogen, a mixture of zinc and chromium oxide is used as a catalyst.

$$CO + 2H_2 \xrightarrow[Cr_2O_3]{ZnO} CH_3OH$$

In Bosch process of preparation of H_2, Cr_2O_3 acts as a promoter for catalyst Fe_2O_3.

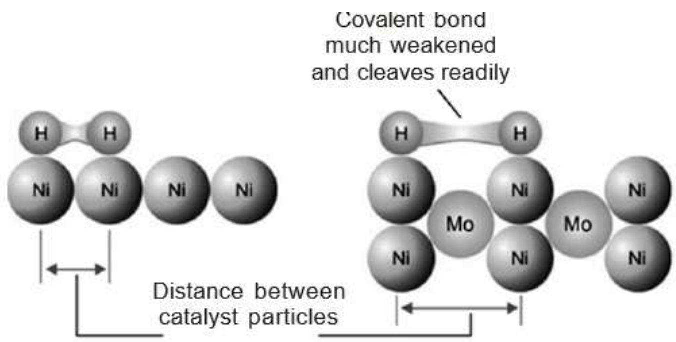
Catalyst Poison or Promoter does no act like a catalyst

Explanation of Promotion Action

The theory of promotion of a catalyst is not clearly understood. Presumably:

- ✓ **Change of Lattice Spacing.** The lattice spacing of the catalyst is changed thus enhancing the spaces between the catalyst particles. The absorbed molecules of the reactant (say H_2) are further weakened and cleaved. This makes are reaction go faster.
- ✓ **Increase of Peaks and Cracks.** The presence of the promoter increases the peaks and cracks on the catalyst surface. This increases the concentration of the reactant molecules and hence the rate of reaction.

The phenomenon of promotion is a common feature of heterogeneous catalysis.



How the change of crystal lattice spacing of catalyst makes the reaction go faster.

<u>Catalytic Poisons</u>

Small amounts of substances can reduce/destroy the activity of catalyst. If the reduction in activity is reversible, the substances are called inhibitors. Inhibitors are sometimes used to increase the selectivity of a catalyst by retarding undesirable reactions.

A substance which destroys the activity of the catalyst to accelerate a reaction is called a poison and the process is called Catalytic Poisoning.

Examples of Catalytic Poisoning

- The platinum catalyst used in the oxidation of sulphur dioxide (Contact Process), is poisoned by arsenic oxide (As_2O_3)
- The iron catalyst used in the synthesis of ammonia (Haber Process) is poisoned by H_2S.
- The platinum catalyst used in the oxidation of hydrogen is poisoned by carbon monoxide

Types of Catalytic Poison

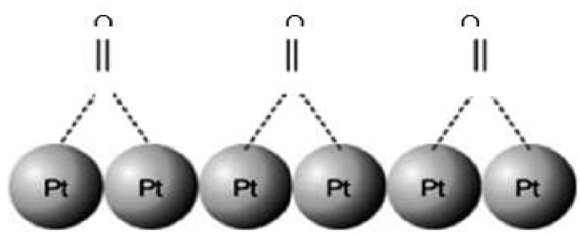
- ✓ Temporary Poisoning

  Catalyst regains its activity when the poison is removed from the reaction
- ✓ Permanent Poisoning

  Catalyst cannot regain its activity even if the catalytic poison is removed. Eg.AS_2O_3 poisons catalyst Pt permanently in manufacturing of SO_3.

Explanation of Catalytic Poisoning

- ✓ The poison is adsorbed on the catalyst surface in preference to the reactants. Even a monomolecular layer renders the surface unavailable for further adsorption of the reactants. The poisoning byAs2O3 or CO appears to be of this kind.



Poisoning of platinum catalyst by carbon monoxide

- ✓ The catalyst may combine chemically with the impurity. The poisoning of iron catalyst by H_2S falls in this class

$$Fe + H_2S \longrightarrow FeS + H_2$$

<u>Acid and Base Catalysis</u>

A number of homogeneous catalytic reactions are known which are catalysed by acids or bases, or both acids and bases. These are often referred to as Acid-Base catalysts. Arrhenius pointed out that acid catalysis was, in fact, brought about by H^+ ions supplied by strong acids, while base catalysis was caused by OH^- ions supplied by strong bases. Many reactions are catalysed by both acids and bases. Typical reactions catalysed by proton transfer are esterification and aldol reaction. Catalysis by either acid or base can be in two different ways

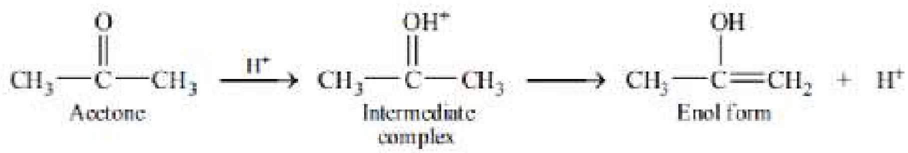
(Specific catalysis and general catalysis)

Example

- ✓ Acid-specific (acid catalysis) — Decomposition of Sucrose into glucose and fructose occurs in presence of sulphuric acid
- ✓ Base-specific (base catalysis) — Addition of hydrogen cyanide to aldehydes and ketones in the presence of sodium hydroxide
- ✓ Inversion of Cane Sugar
- ✓ Keto-Enol Tautomerism of Acetone
- ✓ Decomposition of Nitramide
- ✓ Hydrolysis of an Ester

Mechanism of Acid-base Catalysis

(a) In acid catalysis, **the H+ (or a proton donated by Bronsted acid) forms an intermediate complex with the reactant, which then reacts to give back the proton. For example, the mechanism of keto-enol tautomerism of acetone is:**



(b) In base catalysis, **the OH– ion (or any Bronsted base) accepts a proton from the reactant to form an intermediate complex which then reacts or decomposes to regenerate the OH– (or Bronsted base). For example, the decomposition of nitramide by OH– ions and CH COO– ions may be explained as follows:** ***By*** **OH⁻** ***ions*****:**

$$NH_2NO_2 + OH^- \longrightarrow H_2O + \underset{\text{Intermediate Complex}}{NHNO_2^-}$$

$$NHNO_2^- \longrightarrow N_2O + OH^-$$

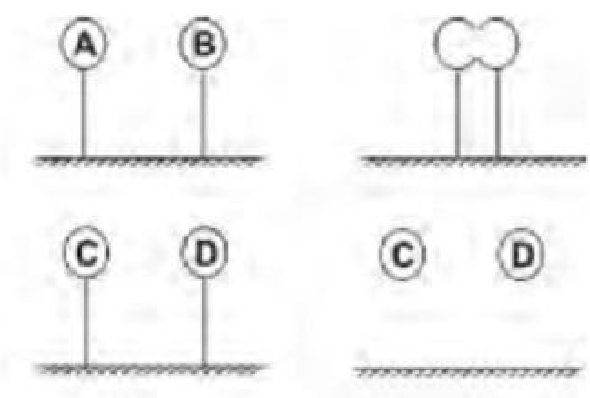


Fig. . . Adsorption

$$A + B \longrightarrow AB$$

presence of a catalyst C, may take place as

$$\underset{}{A} + \underset{\text{Catalyst}}{C} \longrightarrow \underset{\text{Intermediate compound}}{AC}$$

$$AC + B \longrightarrow \underset{\text{Product}}{AB} + \underset{\text{Catalyst}}{C}$$

The Intermediate Compound Formation Theory

MECHANISM OF CATALYTIC REACTIONS

The following two theories have been widely accepted:

(i) Intermediate compound formation theory

(ii) Adsorption theory

(i) Intermediate compound formation theory

This theory explains the catalytic reaction due to the formation of an unstable intermediate compound which is more reactive compared to the reactants. The following examples can be given in support of the above theory:

- ✓ NO_2 acts as an oxygen carrier in Lead Chamber process.

$$2NO + O_2 = 2NO_2 \text{ ... fast}$$
$$SO_2 + NO_2 = SO_3 + NO$$

the net reaction being

$$2SO_2 + O_2 = 2SO_3 \text{ (slow if uncatalysed)}$$

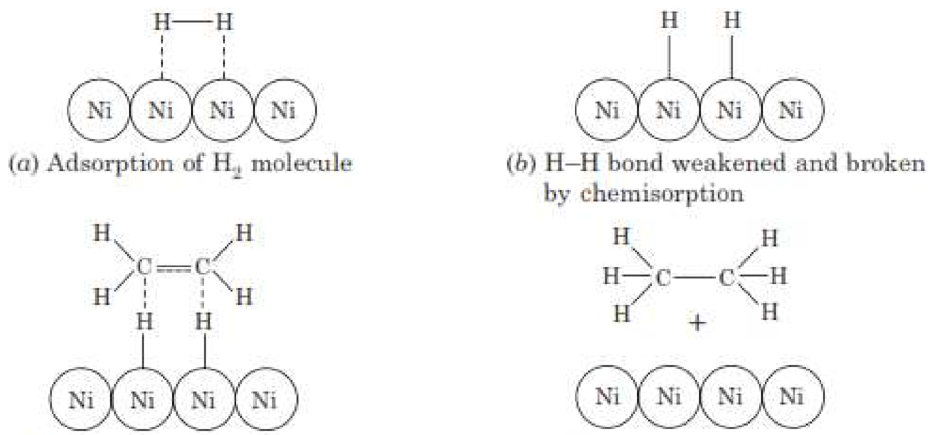
- ✓ The intermediate formation of ethyl hydrogen sulphate in the dehydration of alcohol by conc. H_2SO_4 to ether.
- ✓ In Friedel—Craft reaction, anhydrous $AlCl_3$ forms an intermediate compound

with RCl and finally generates R^+ for the reaction.

$$RCl + AlCl_3 \longrightarrow \overset{\delta+}{R} \ldots \overset{\delta-}{AlCl_4} \rightleftharpoons R^+ + AlCl_4^-$$

(ii) Adsorption theory (Heterogeneous catalysis)

This theory mainly explains the mechanism of a catalytic reaction between two gasescatalyzed by a solid. Here, the catalyst works by adsorption of the gas molecule on its surface. The adsorption theory is compared to the intermediate compound formation theory. Here, the intermediate compound is an activated complex formed on the catalyst surface. Theadsorption theory can be explained by the following Fig.



(*a*) Adsorption of H_2 molecule

(*b*) H–H bond weakened and broken by chemisorption

(*c*) Formation of activated complex with ethene.

(*d*) The product ethane released from catalyst surface.

APPLICATION OF CATALYSIS

1. Catalytic Converters

A device incorporated in the exhaust system of a motor vehicle, containing a catalyst for converting pollutant gases into less harmful ones.

Catalytic converters are used for mitigating automobile exhaust emissions. Catalytic converters are used within internal combustion engines fueled by either petrol (gasoline) or diesel. It converts three harmful substances into harmless ones carbon monoxide (a poisonous gas) into carbon dioxide, nitrogen oxides (cause acid rain and smog) into nitrogen and oxygen, and hydrocarbons (cause smog and respiratory problems) into carbon dioxide and water.

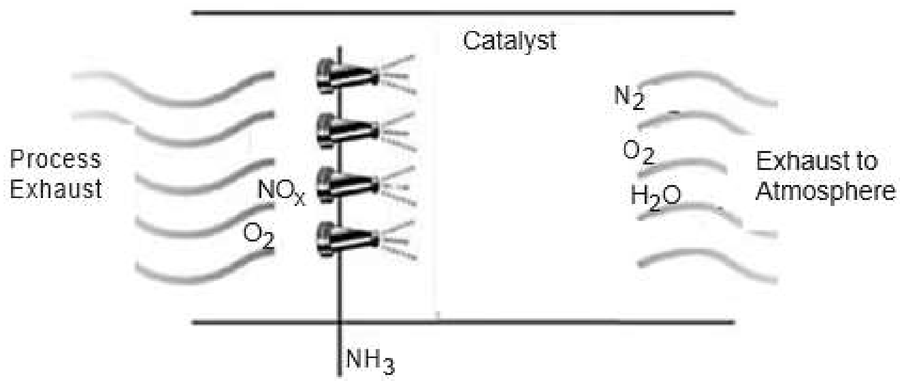
Catalytic converters consist of a stainless steel box attached to the muffler and containing ceramic beads or honeycomb coated with catalysts (Expensive metals like platinum, Palladium, Rhodium are used. Alumina, Ceria can also be used when combined with expensive metals.)

Eg: Consider the reaction

$$2\,CO + 2\,NO \longrightarrow 2\,CO_2 + N_2$$

Carbon monoxide and Nitrogen monoxide will be adsorbed on the surface of catalyst where they get adsorbed to catalyst and result in formation of carbon dioxide and nitrogen which gets desorbed. Catalytic converters can be affected by catalytic poison.

Eg. Lead is a very good example for catalytic poisoning. It gets adsorbed to the honey comb of expensive metals and inhibits the function of catalyst. Catalytic converter has also forced the removal of lead from petrol.



$$4\,NO + 4\,NH_3 + O_2 \longrightarrow 4\,N_2 + 6H_2O$$

$$2\,NO_2 + 4\,NH_3 + O_2 \longrightarrow 3\,N_2 + 6H_2O$$

Basic Catalytic Converter

2. Petroleum Refining

(i) ***Fluid catalytic cracking***: Breaking large hydrocarbon into smaller hydrocarbons.

(ii) ***Catalytic reforming***: Reforming crude oil to produce high quality gasoline component

(iii) ***Hydrodesulfurization***: Removing sulphur compounds from refinery intermediate products

(iv) ***Hydrocracking***: Breaking large hydrocarbon molecules into smaller ones

(v) ***Alkylation:*** Converting isobutane and butylenes into a high-quality gasoline component

(vi) ***Isomerization:*** Converting pentane into a high-quality gasoline component

3. Chemicals and petrochemicals

(i) Haber process for ammonia production

(ii) Styrene and Butadiene synthesis for use in producing synthetic rubber

(iii) Contact process for production of sulphuric acid

(iv) Ostwald process for production of nitric acid

(v) Methanol synthesis

(vi) Production of different plastics and synthetic fabrics

4. Other

(i) Fischer-Tropsch and Coal gasification processes for producing synthetic fuel gases and liquid fuels

(ii) Various processes for producing many different medicine

Catalysis for a Better Tomorrow

Catalysts will play a pivotal role in achieving a greener future. They can help produce value-added chemicals for renewable energy, capture carbon dioxide from the air, and provide clean water, among other things. This will help our nation achieve its energy and sustainability goals.